IT Italiano · GB **English**

🔧 ichosynth — Build Manual

DIY hand-wired build (no printed PCB)

Step-by-step guide for beginners: build your own **ichosynth** using nothing but flying wires (jumpers), following the pin tables.

Difficulty Medium (SMD PSRAM) | Build 4 encoders + 3 buttons | runs TŒRN · SP_ | See also Usage Manual

> 🧠 **ichosynth** is a hand-soldered, low-cost build of **TŒRN** (by SP_ / soundpauli, https://toern.live): an open-source sampler-sequencer based on the **Teensy 4.1**. It runs the **real TŒRN firmware** — ported to cheap, solderable input parts — and generates all its sounds on its own; the computer is needed **only** to program it the first time.

> 🆕 **What changes vs. TŒRN.** TŒRN's expensive parts (I^2C RGB encoders, capacitive touch pads) are swapped for **4 cheap KY-040 rotary encoders** (turn + push) and **3 tact switches**. TŒRN's per-encoder RGB ring feedback is replaced by a small **SSD1306 OLED** that shows channel, mode, transport, BPM, volume and page. Everything else — the DSP, the sequencer, the sampler — is the unchanged TŒRN firmware.

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1 · What you are building

- A brain: the **Teensy 4.1** (a powerful microcontroller).
- An audio board (**Teensy Audio Adaptor**, SGTL5000) with a 3.5 mm headphone output.

- A playing display: a **16×16 RGB LED matrix** (256 LEDs, chainable to 32×16).
- **Four rotary knobs with push-buttons** (KY-040 encoders): E1 (left) → E4 (right).
- **Three tact switches**: PLAY, MENU, REC.
- A small **OLED screen** (SSD1306 128×64) showing channel, mode, transport, BPM, volume, page.
- A **micro SD** card that holds the audio samples and your patterns.

Everything is powered from the **USB (5V)** port.

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2 · Difficulty level — read before you buy

> ⚠️ **Honesty first.** The wiring is easy; the hard part is just **one thing**: soldering the two SMD PSRAM chips onto the back of the Teensy.

Part	Difficulty	Notes
Matrix + encoder + button wiring	🟢 Easy	chunky solder joints and wire connections, suitable for patient beginners
PSRAM soldering (2× SMD chips)	🔴 Hard	tiny components; see options below

- **PSRAM — option A (recommended):** buy the Teensy 4.1 **with the PSRAM already soldered**, or have it soldered by someone with experience / a hot-air station.
- **PSRAM — option B:** practice first on practice boards.
- ⚠️ The PSRAM **is mandatory**: TŒRN uses ~16.5 MB of external memory, so you need **both** 8 MB chips soldered (16 MB total). Without them the firmware will not run.

> ⏱️ Estimated time: half a day for someone who already knows how to solder; longer if it's your first time.

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3 · Bill of materials (BOM)

Qty	Component	Notes
1	Teensy 4.1	the main microcontroller
2	8 MB PSRAM chip (APS6404, Teensy 4.1 compatible)	16 MB total, **mandatory (both chips)**
1	Teensy Audio Adaptor Board, **Rev D (for Teensy 4.x)**	SGTL5000 codec + 3.5 mm headphone jack (no speaker); its SD slot is not used
1	**WS2812 RGB 16×16** LED matrix (256 LEDs)	rigid or flexible; chainable to 32×16
4	**KY-040** rotary encoder with push-button	E1 (left) → E4 (right)
3	**6×6 mm** tact switch (momentary; **4.3 mm tall** if you use the case)	PLAY · MENU · REC (each: one leg to pin, other leg to GND)

Qty	Component	Notes
1	Micro SD Card, **Class 10**	for samples and patterns
1	Micro-USB cable + 5V power supply (≥ 2A recommended)	power and programming
1	Headphones with 3.5 mm jack	ichosynth has no speakers; on-board jack for monitoring
1	**6.35 mm (1/4") mono TS jack**	Line In (mono, for recording) → Audio Shield LINE IN L pad
as needed	*(optional)* 10 µF electrolytic capacitor	series DC-blocking on the line jacks, only if connected gear hums
1	**SSD1306 0.96" 128×64 I²C** OLED	status screen (replaces TŒRN's encoder RGB rings)
as needed	AWG 26 wire in 6+ colours (~10-15 cm per wire)	for the (soldered) connections
2	**Long male pin header** strips (≥ 15 mm, 14 pins)	to stack the audio board on the Teensy
2	*(recommended)* 5-pole lever connectors (e.g. WAGO 221-415)	3.3 V and GND bus bars, no soldering
1	*(optional)* **3D-printed case** (3 parts, 2 plates 250×250)	`_CASE/` — graphic guide `_CASE/MONTAGGIO.pdf` ; also needs 8 M3×8 countersunk screws and 4 feet

> ℹ️ **Samples**: the factory set in `_SDCARD/` is **public domain** (the *drum_sklad* collection), freely redistributable. You can add your own (see ch. 9).

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4 · Tools you'll need

- 🔥 Fine-tip soldering iron + solder (and flux, very helpful for the PSRAM).
- ✂️ Side cutters, wire strippers, tweezers.
- 🤚 A "third hand" or vise to hold the parts steady.
- 📟 Multimeter (continuity and short-circuit checking — **essential**).
- 🧴 Isopropyl alcohol to clean up flux residue.
- ⚡ Anti-static precautions (ESD wrist strap): the Teensy and PSRAM are sensitive.
- *(only if you solder the PSRAM yourself)* a hot-air station or a precision soldering iron.

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5 · Basic safety concepts

> ⚠️ Four rules that save your components (and your nerves):

1. **Never** connect/disconnect wires while the device is powered.

2. Double-check **GND and 5V/3.3V** before applying power: swapping them can fry your components.
3. After every batch of solder joints, use the multimeter in continuity mode to verify there are **no** shorts between 5V and GND.
4. Work calmly: a "cold" solder joint (dull, blobby) is the #1 cause of malfunctions.

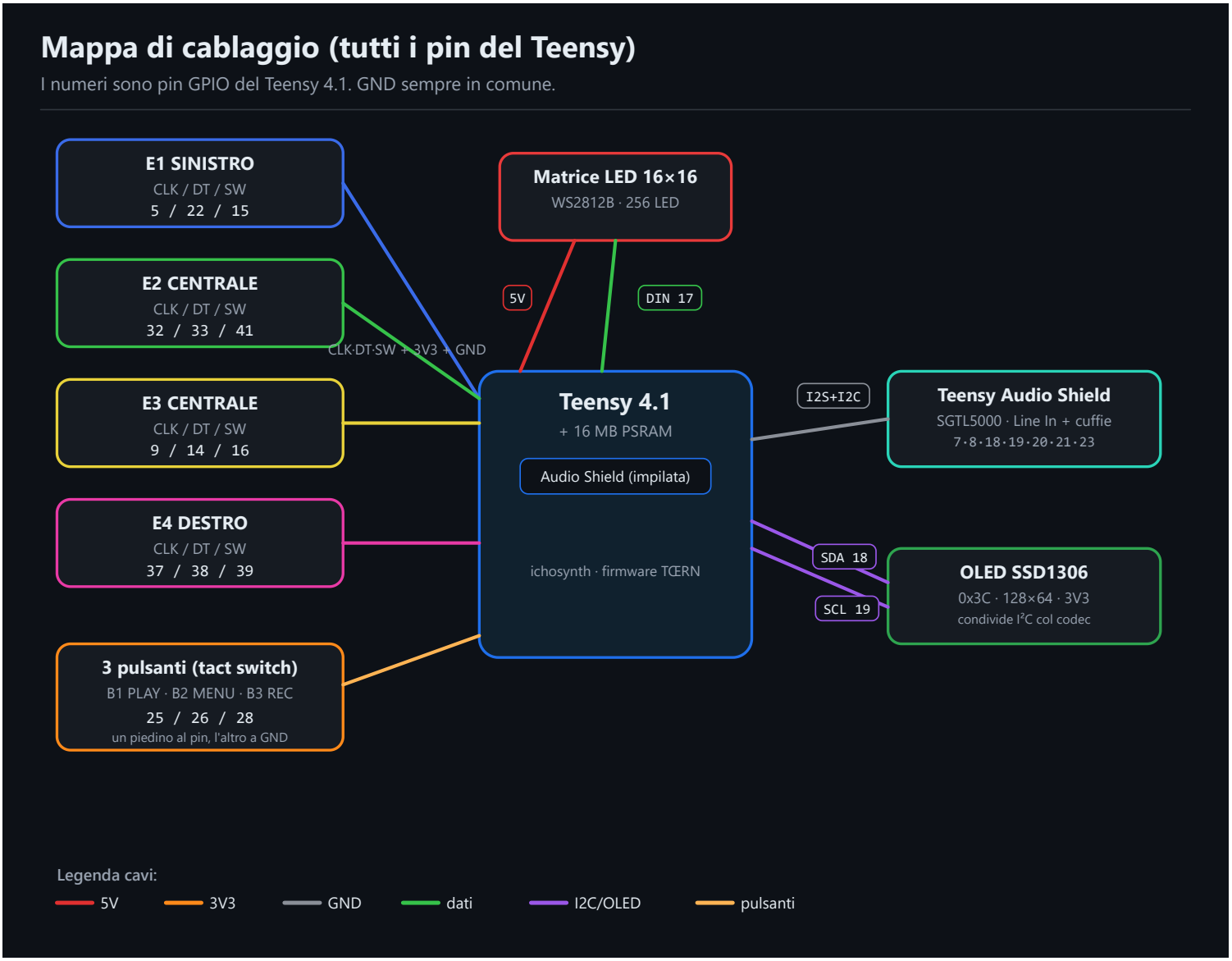
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6 · Complete pin map (the firmware's "truth")

> 🎨 **Illustrated wiring guide:** `_DOCS/CABLAGGIO.pdf` (Italian) — 7 plates with the real parts drawn (true Teensy 4.1 pinout, KY-040, OLED, buttons, matrix, jack), one wire at a time, wire colours and a final check. The pin numbers are read straight from the firmware drivers. This chapter stays as the tabular reference.

These pins are what the firmware expects. They live in the build sources — the encoder/button tables `ICHOS_ENC_PINS` and `ICHOS_BTN_PINS` under `teensy/libraries/`, plus `teensy/build_toern.py` — **not** in a `config.h`. **Wire exactly these numbers** (all of them are Teensy pins).

> 🧭 These pins were chosen deliberately to avoid TŒRN's hard-coded GPIO and the Teensy Audio Shield's reserved pins. **Do not use pins 2 / 3 / 4** (TŒRN internal use) or **pin 24** (was an optional LED strip, removed in this build).



6.1 LED matrix

Matrix signal	Teensy pin
DIN (data)	**17**
+5V	**5V**
GND	**GND**

6.2 Audio board (Teensy Audio Adaptor, Rev D)

> 💡 The simplest and most reliable way is to **stack** the audio board on top of the Teensy using **long male pin headers** (headphone jack on the same side as the USB): these connections make themselves, and the wires for the pins covered by the audio board are soldered onto the pin tips sticking out above it (see `_DOCS/CABLAGGIO.pdf`, plate 3).

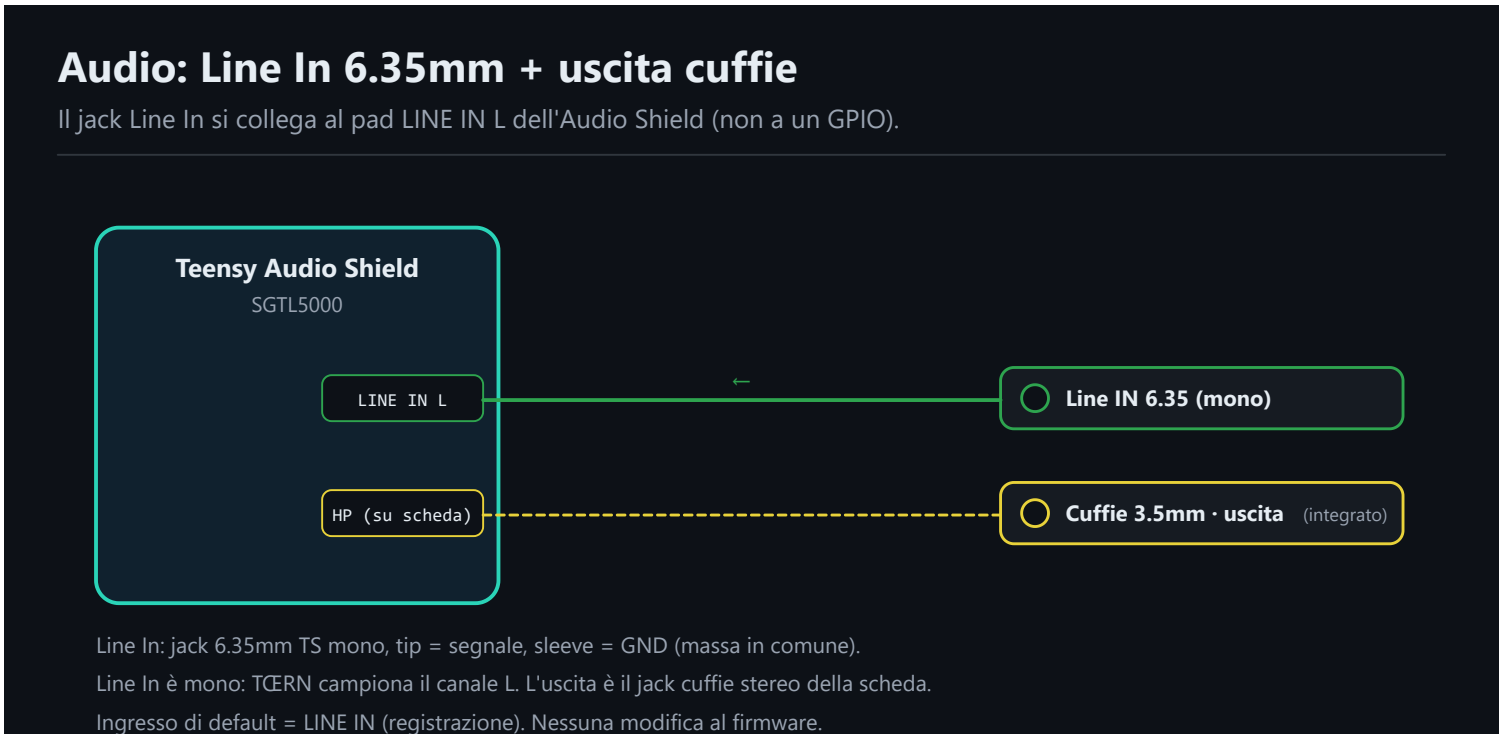
If instead you wire it by hand, connect:

Audio signal	Teensy pin		Audio signal	Teensy pin
MCLK	**23**		SDA (I^2C)	**18**
BCLK	**21**		SCL (I^2C)	**19**
LRCLK (WS)	**20**		3.3V	**3.3V**
TX (DIN to codec)	**7**		GND	**GND**
RX (DOUT from codec)	**8**			

> ℹ️ The SD is used from the **Teensy 4.1's built-in slot**, not the one on the audio board.

6.2.1 Audio: Line In + headphones

The output is the audio board's **on-board 3.5 mm headphone jack** (stereo). To record from outside gear you add **one 6.35 mm (1/4") mono TS jack for Line In**, wired to the **LINE IN L solder pad on the Teensy Audio Adaptor** — **not** to a Teensy GPIO pin.



Jack	Tip (signal) → Audio Shield pad	Sleeve
Line In 6.35 mm mono TS	**LINE IN L**	**GND** (common ground)
Headphones 3.5 mm	*(on-board jack)*	—

- **Line In is mono.** TŒRN records/monitors the **LEFT** line-in channel as a mono sample, and LINE IN is already the firmware's default recording input — so **no firmware change is needed**.
- *(Optional)* If the connected source hums, add a **10 µF series DC-blocking electrolytic cap** in line with the signal. Direct wiring is normally fine.

> 🎙️ **What they're for: Line In** = sample external instruments / a field recorder / other gear straight into the sampler. **Headphones** (3.5 mm) = listening and output.

6.3 Encoders (CLK, DT, SW = push-button)

You fit **4 encoders**. Each encoder has 3 signals (CLK, DT, SW) + power. Arrange them from left to right (E1 → E4) as shown in the table.

Encoder	CLK	DT	SW
E1 (left)	**5**	**22**	**15**
E2	**32**	**33**	**41**
E3	**9**	**14**	**16**
E4 (right)	**37**	**38**	**39**

In addition, for each encoder: the **"+"** pin goes to **3.3V**, the **GND** pin goes to **GND**.

6.4 Buttons — 3 tact switches (PLAY / MENU / REC)

Plain momentary pushbuttons (tactile/push type). Each one: **one leg to the Teensy pin, the other leg to GND**. They use the Teensy's **internal pull-up** (active-low) — no external resistor needed.

Button	Role	Teensy pin
B1	PLAY	**25**
B2	MENU	**26**
B3	REC	**28**

6.5 OLED (SSD1306)

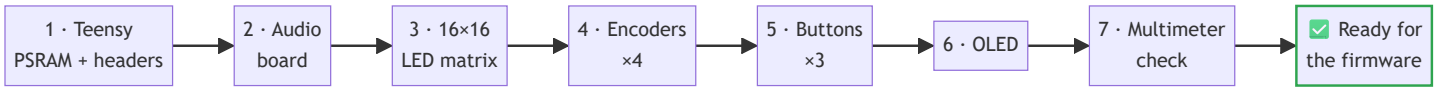
It shares the **same I²C bus as the audio codec** (no extra signal pins: same SDA/SCL). It also needs its own 3V3 and GND.

OLED signal	Teensy pin
SDA	**18**
SCL	**19**
VCC	**3.3V**
GND	**GND**

> ℹ️ Default I²C address **0x3C** (some panels use 0x3D). The OLED replaces TŒRN's encoder RGB-ring feedback, showing channel / mode / transport / BPM / volume / page.

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7 · Step-by-step assembly



Step 1 — Prepare the Teensy 4.1

1. Solder the **two PSRAM chips** onto the pads on the back (see ch. 2: if you don't feel up to it, get a Teensy with the PSRAM already fitted). Both chips are mandatory.
2. Solder the **headers** onto the edges of the Teensy (and those for the audio board if you stack it).
3. Connect it to the PC via USB and check that it's recognized (full test in Step 8 / ch. 10).

Step 2 — Audio board

1. Stack the audio board on the Teensy (recommended) **or** wire the signals from table 6.2.
2. Plug the headphones into the 3.5 mm jack (for testing).

Step 3 — 16×16 LED matrix

1. Find the **data input arrow** (input): it goes to pin **17**. The output stays free (chainable to 32×16 if you add a second panel).
2. Connect the matrix's **5V** and **GND**.
3. ⚡ **Power**: the firmware uses low-brightness colors, so USB is usually enough. If you raise the brightness in the future, inject the 5V from a dedicated power supply and **join the grounds (common GND)** between the Teensy and the matrix.

Step 4 — Encoders (×4)

1. For each of the **4 encoders**, connect CLK, DT, SW per table 6.3, plus "+" (3.3V) and GND.
2. Arrange them from left to right: **E1** → **E2** → **E3** → **E4** (with the 3D-printed case they sit 2×2 instead: see 3D-printed case).
3. Keep the wires tidy and label them: crossing CLK/DT is the most common mistake (it can also be fixed in software, see troubleshooting).
4. ⚠️ Don't accidentally use pins **2 / 3 / 4** (TŒRN internal use) or **24** (removed LED strip) — none of the encoders touch them.

Step 5 — Buttons (×3)

1. Wire each tact switch with **one leg to its Teensy pin** and **the other leg to GND** per table 6.4.
2. No resistors needed: the firmware enables the internal pull-ups (active-low).

Step 6 — OLED

1. Connect the 4 wires from table 6.5. Since it's on the same bus as the audio, you just connect it in parallel to SDA/SCL; give it its own 3V3 and GND.

Step 7 — Final check before powering on

1. With the multimeter, verify there's **no short** between 5V and GND and between 3.3V and GND.
2. Double-check that 5V/3.3V/GND are on the correct pins.

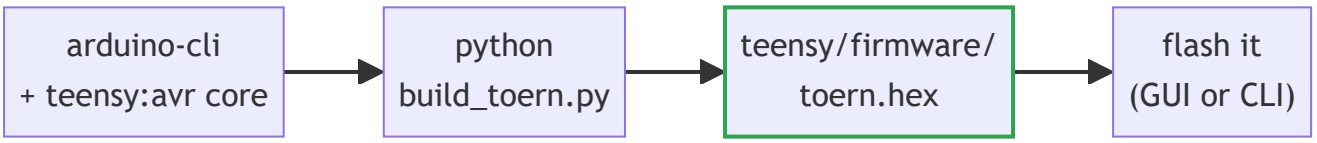
3D-printed case (optional)

The 3D-printed case lives in `_CASE/`: 3 parts on 2 print plates, no supports. The **fully graphic** step-by-step guide is `_CASE/MONTAGGIO.pdf`; printing and parameters are in `_CASE/README.md`. With the case the encoders sit 2×2 (E2 E3 top, E1 E4 bottom: the Y/X cursor knobs nearest the hand), wires are soldered straight onto the KY-040 pins, the tact switches must be the **4.3 mm tall** ones and headphones exit at the back. No Dupont connectors inside the case: wires are soldered (on the Teensy from the top side), and the 6.35 mm Line In jack must fit in **22 mm of depth and 16 mm of height** behind the wall.

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8 · Software: flashing the firmware

The firmware is the **real TŒRN** ported to our hardware. You don't edit Arduino settings by hand — a build script handles the whole toolchain (it builds at `-O1`, the option TŒRN's large single translation unit needs to compile cleanly on the Teensy gcc).



8.1 Build from source (recommended)

1. Install **arduino-cli** and add the **teensy:avr** core (≥ 1.61.0).
2. From the repo root, run:

   ```
   python teensy/build_toern.py
   ```

   This produces **`teensy/firmware/toern.hex`**. See `teensy/README.md` for details (it also clones the TŒRN sources if missing and applies the pin remap for our 3 buttons).
3. To build **and** flash in one go (needs `teensy_loader_cli`):

   ```
   python teensy/build_toern.py --flash
   ```

> 🧱 The build pins the optimization level to **`opt=o1std` (-O1)**. At the default `-O2` the Teensy gcc can crash on TŒRN's ~23k-line single file; `-O1` compiles cleanly and still fits with room to spare.

8.2 One-click GUI flasher (prebuilt .hex)

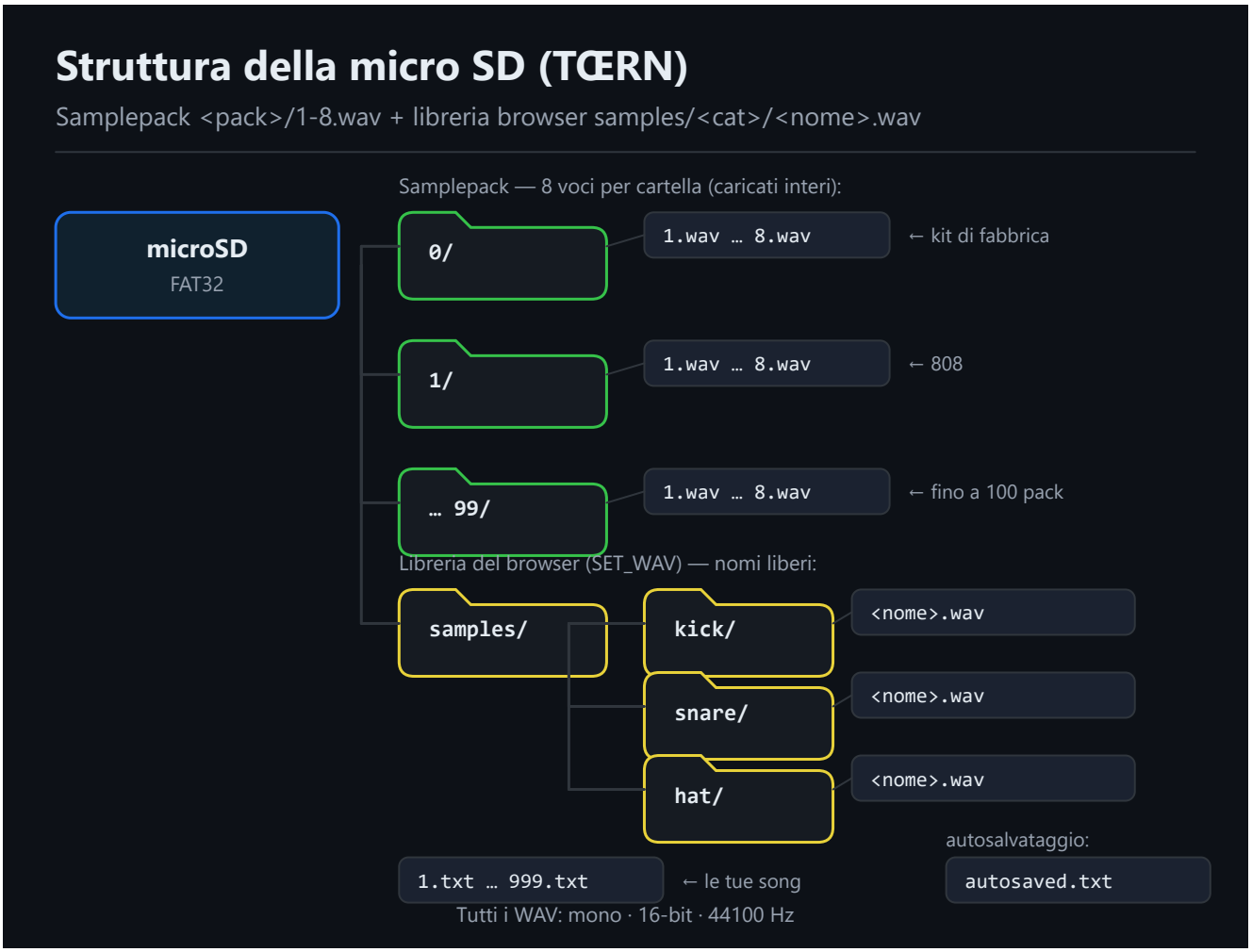
If you already have a `toern.hex`, the **`_FLASHER`** folder contains a one-click GUI flasher: pick the board, pick the `.hex`, press flash. No Arduino IDE required.

> ℹ️ Hardware note: this is a **Teensy 4.1**. The build script sets the USB type for you (Serial + MIDI). If the loader doesn't see the board, press the little button on the Teensy to enter programming mode.

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9 · Preparing the micro SD (samples)

The firmware reads samples from the SD with **two** systems:



/1-8.wav and browser library samples//.wav" width="620">

1 · Samplepacks — a numbered folder `0` .. `99` on the **root**, each containing exactly **8 voices** `1.wav` .. `8.wav` (one per sample channel). The firmware loads a whole pack at boot and when you pick one from the menu.

```
/0/1.wav .. /0/8.wav     (pack 0 = factory default kit)
/1/1.wav .. /1/8.wav     (pack 1)
/2/1.wav .. /2/8.wav     …
```

2 · Browser library — free WAV files under `samples/<category>/<name>.wav`, browsed voice-by-voice in **SET_WAV** (up to 256 files per folder, names up to 60 characters).

```
/samples/kick/...   /samples/snare/...   /samples/hat/...   /samples/perc/...
```

> ⚠️ **Required audio format (both systems): WAV mono, 16-bit, 44100 Hz.** The old NI404 naming (`samples/0/_1.wav` , `_<n>.wav`) is **not** read by the TŒRN firmware.

The project's `_SDCARD/` folder already ships a ready-made set (6 packs + a browser library); just copy its contents to the root of the card. See `_SDCARD/README.md` .

Converting your samples

In `_SDCARD/` you'll find the tools that convert any WAV into the right format and place it.

🪟 `wavmaker.exe` **— graphical interface (Windows, no Python required), recommended**

Double-click: a window opens. Then:

1. Pick a **mode** at the top:
   - **Samplepack (8 voices)** — the files become `<pack>/1.wav … 8.wav` .
   - **Browser library** — the files go to `samples/<category>/<name>.wav` (names kept).
2. **Add files** (or **Add folder**) with your WAVs — the list shows each file's **current format** and its **destination**. **Green** rows are already mono/16-bit/44100. In samplepack mode, drag with the ↑ ↓ buttons to set which file is which voice.
3. **Scan** the SD (it detects existing packs and the next free pack number), or tick "*Write to a local folder*" to prepare the card offline.
4. Pick the **pack number** (or the **category**) and press **Convert**.

> ✅ Non-destructive by default — originals are kept unless you tick "*Delete originals*".

🐍 **Python versions** (Windows/macOS/Linux): `python wavmaker_gui.py` (same GUI) or `python wavmaker.py` (command line). No `audioop` /ffmpeg needed — the converter is built in, so it works on Python 3.13/3.14 too.

> 🧱 To regenerate the whole factory set from public-domain sources: `python _SDCARD/build_samples.py` .

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10 · First boot and testing

1. Insert the SD, plug in the headphones, power it via USB.
2. On power-up you'll see an **animation/logo** on the matrix, and the **OLED** lights up with the status (channel, mode, BPM, volume, page).
3. If the SD is missing, the **"noSD"** icon (red) appears: power off, insert the SD, power back on.
4. Move the encoders: the matrix and the OLED should react (cursor, volume, BPM, ...).
5. Press an encoder / a button to place a note or trigger transport: you should hear the sample.
6. Try the **PLAY** button to start/stop the sequencer.

> 🎮 To learn how to play it, head to the **Usage Manual** and the control map in `_DOCS/MAPPA_CONTROLLI.md`.

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11 · Troubleshooting

Symptom	Likely cause / fix
🕯️ Won't power on / not recognized by the PC	Charge-only USB cable (use a data one); header solder joints; 5V-GND short
🔁 Reboots on its own / crashes after a few seconds	PSRAM missing or badly soldered (both chips required); weak USB power (use 5V ≥ 2A)
💡 LEDs don't light up / wrong colors	DIN not on pin 17; data arrow reversed (use the input); GND not common
🌫️ Only some LEDs light up randomly	Insufficient power to the matrix; missing common GND
↩️ An encoder turns "backwards"	Swap the **CLK and DT** wires on that encoder
🔇 An encoder does nothing	Signals on the wrong pins; recheck table 6.3; make sure you didn't land on 2/3/4
🔘 A button does nothing	One leg must go to the pin, the other to **GND**; recheck table 6.4 (25/26/28)
🎧 No sound in the headphones	Audio board not connected properly (7,8,20,21,23,18,19); volume at 0
❌ Build fails / compiler error	Make sure `arduino-cli` and the `teensy:avr` core are installed; run `python teensy/build_toern.py` (it builds at -O1 — see `teensy/README.md`)
🚫 Samples won't play / channel muted	Wrong SD structure/path; WAV not mono-16bit-44.1k (use `wavmaker.py`)
📟 OLED black	Address 0x3D instead of 0x3C; SDA/SCL swapped; missing 3V3/GND

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12 · Pin summary (cheat-sheet)

```
LED matrix DIN ............ 17        Audio MCLK ... 23   Audio SDA ... 18
                                     Audio BCLK ... 21   Audio SCL ... 19
Encoder E1 (left)  CLK 5  DT 22 SW 15 Audio LRCLK .. 20
Encoder E2         CLK 32 DT 33 SW 41 Audio TX ..... 7
Encoder E3         CLK 9  DT 14 SW 16 Audio RX ..... 8
Encoder E4 (right) CLK 37 DT 38 SW 39 OLED ......... SDA 18 / SCL 19 (0x3C)
Buttons  PLAY 25   MENU 26   REC 28   (each: pin -> GND, active-low)
Avoid:   pins 2/3/4 (TŒRN internal) and 24 (removed LED strip)
```

> ⚡ Power rails: matrix = **5V**; encoders, OLED and audio = **3.3V**; **GND always common** across everything.

Happy building! 🛠️ Once it makes sound, move on to the **Usage Manual**.

*ichosynth is a hand-soldered, low-cost build of **TŒRN** by SP_ (soundpauli, https://toern.live) · open-source firmware running on a Teensy 4.1.*